

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ S0/0/0 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ____ 129 ____

Does R2 show up as an OSPF neighbor on R1? ____ Yes ____

Does R2 show up as an OSPF neighbor on R3? ____ No ____

What does this information tell you?

____ The way from R3 to 192.168.2.0/24 network will cross R1. _____

____ The S0/0/1 interface on R2 still works, so OSPF routing information will not cross this interface. _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ S0/0/1 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____ 65 (64 of T1 link+1 of R2 link). _____

Is R2 listed as an OSPF neighbor to R3? ____ Yes ____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The OSPF default reference-bandwidth has low transmission rate, so cost is lower than today's devices. So a higher default reference-bandwidth setting is needed.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24: $R1(S/0/0/1) + R3(G0/0) = 781 + 1 = 782$

Cost to 192.168.23.0/30: $R1(S/0/0/1) + R3(S/0/0/1) = 781 + 64 = 845$

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Route to 192.68.23.0/24 network cross over 2 serial links with cost 781, $781 \times 2 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF choose the lowest accumulated cost route. The $R1-S0/0/0 + R2-S0/0/1 + R3-G0/0 = 781 \times 2 + 1 = 1563$, and is lower than $R1-S0/0/1 + R3-G0/0 (1565 + 1 = 1566)$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

In OSPF, the router ID controls Designated router and backup designated router election. It can change if the interface goes down, so it needs to be set.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR is for multi-access network, but this lab is point to point.

3. Why would you want to set an OSPF interface to passive?

It can reduce useless OSPF routing information on that interface, make bandwidth more space.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The areas makes OSPF hierarchical and scalable. This allows route summarization between areas and limits the propagation of link-state updates within a single area

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___ Routers use administrative distance (AD) to choose between routes learned from different routing protocols, not the metric values. The route with the lowest administrative distance is installed in the routing table.

Typical default administrative distances:

RIP: 120

OSPF: 110

EIGRP: 90 (internal)

Since OSPF has an AD of 110, which is lower than RIP's 120, it is preferred over RIP. However, EIGRP has an even lower AD of 90, the result is as below:

RIP AD = 120

OSPF AD = 110

EIGRP AD = 90 (internal) / 170 (external)___

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